

Build, Release, and Containerize an Application

Jeffrey Gregory

SDI 4213-980 - DevOps: CI/CD
Weeks 5 and 6 Individual Exercise

15 / 15	v0.1.0	0.1.0	HTTP 200
Tests passed	GitHub Release	Docker image tag	Health check

Repository

https://github.com/jwm-dev/SDI_4213_Weeks_5_6

Completed work

The supplied FastAPI inventory application and all 15 tests were preserved. CI now creates and retains a release ZIP after tests pass. The project has a versioned Git tag and GitHub Release, a Dockerfile and .dockerignore, and a verified Windows Docker image and container.

GitHub issue and pull request record

- Issue #1: Add automated build artifact
- PR #2: Add automated build artifact - merged, CI passed
- Issue #3: Containerize application
- PR #4: Containerize application with Docker - merged, CI passed

Evidence timing and assistance

GitHub implementation milestones occurred October 1, 2026. This document records Windows verification and screenshots captured October 6, 2026 (America/Chicago). GitHub screenshots show those historic runs and the published release. Milestone 1 revalidates unmodified starter commit 333bc5f in a separate checkout; it is not a claim that the screenshot was taken before original development.

Prepared for Jeffrey Gregory with Codex assistance in review, tool execution, Docker troubleshooting, screenshot capture, and document preparation. PR reviews are documented self-review comments; they are not independent approvals.

Verification details

Starting project and CI

Local Python 3.13.14: 15 starter tests pass in the separate starter checkout; 15 tests also pass in the final project. The original application source, pinned requirements, and tests remain unchanged.

Workflow CI runs for pull requests targeting main and pushes to main on ubuntu-latest. It checks out the repository, configures Python 3.13 using actions/setup-python@v5 with pip caching, installs requirements with python -m pip, and runs python -m pytest -v.

Baseline: [successful run 36925881257](#).

Build artifact and release

After tests pass, CI copies app/, requirements.txt, README.md, and VERSION into dist/package and creates dist/sdi4213-app.zip. actions/upload-artifact@v4 uploads sdi4213-app with 90-day retention. The downloaded ZIP passed integrity and source-file equality checks; extracted required files and VERSION (0.1.0) were verified.

Build PR: [#2](#); build run: [36926050064](#).

Release: [v0.1.0 - Version 0.1.0 - Build and Containerization Exercise](#). The tag points to build merge commit 3ce31ba. Following the assignment order, the release tag precedes the Docker PR. Container configuration is subsequently merged into main; application source and VERSION remain unchanged.

Windows Docker verification

Docker Desktop 4.53.0, Engine 29.0.1, Compose v2.40.3-desktop.1, context desktop-linux. docker run --rm hello-world completed successfully. Terminal screenshots show an xterm viewer backed by real Windows PowerShell and actual command output.

Dockerfile uses python:3.13-slim, /app, dependency-first COPY/install, port 8000, and Uvicorn at 0.0.0.0:8000.

.dockerignore excludes the assignment-required Git, virtual environment, cache, bytecode, .env, and dist files.

Built sdi4213-week56:0.1.0 with container Python 3.13.16; container 2fe4e125cf06 ran as sdi4213-week56-demo with 8000:8000 published. /health returned status ok and HTTP 200; logs show startup and successful requests. The first request raced startup, then verification completed.

Full Windows image ID:

sha256:1cf7aad2feb08b59f5205f116655ea8d1664d61e45551256037039c61adee14

Cleanup and final checks

The demonstration container was stopped and removed. Image inspection after removal confirmed that the same versioned image and full image ID were retained. Windows command outputs are retained in docs/evidence/windows/.

Final Docker PR CI: [successful run 36926836443](#). The self-review and passing checks are recorded on [PR #4](#).

01 Local Tests

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 1 • Completed • exit 0

Captured session: 2026-10-06 09:06:39 -05:00

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\starter-validation> git log -1 --oneline
333bc5f (HEAD) Add Weeks 5-6 starter project
```

Unmodified starter commit 333bc5f; revalidated on Windows October 6, 2026.

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\starter-validation> python --version
Python 3.13.14
```

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\starter-validation> python -m pytest -v
===== test session starts =====
platform win32 -- Python 3.13.14, pytest-8.4.1, pluggy-1.6.0 -- C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\starter-validation
plugins: anyio-4.15.1
collected 15 items

tests/test_api.py::test_root_route_returns_message PASSED [ 6%]
tests/test_api.py::test_health_check_returns_ok PASSED [ 13%]
tests/test_api.py::test_read_items_returns_count_and_items PASSED [ 20%]
tests/test_api.py::test_read_item_returns_one_item PASSED [ 26%]
tests/test_api.py::test_read_item_returns_404_for_missing_item PASSED [ 33%]
tests/test_api.py::test_add_item_creates_item PASSED [ 40%]
tests/test_api.py::test_add_item_rejects_duplicate_id PASSED [ 46%]
tests/test_services.py::test_get_all_items_returns_starter_items PASSED [ 53%]
tests/test_services.py::test_get_item_by_id_returns_matching_item PASSED [ 60%]
tests/test_services.py::test_get_item_by_id_returns_none_for_missing_item PASSED [ 66%]
tests/test_services.py::test_calculate_total_quantity PASSED [ 73%]
tests/test_services.py::test_is_low_stock_true_when_quantity_at_threshold PASSED [ 80%]
tests/test_services.py::test_is_low_stock_false_when_quantity_above_threshold PASSED [ 86%]
tests/test_services.py::test_create_item_adds_new_item PASSED [ 93%]
tests/test_services.py::test_create_item_rejects_duplicate_id PASSED [100%]
```

```
===== warnings summary =====
..\SDI_4213_Weeks_5_6\.venv\Lib\site-packages\starlette\testclient.py:51
C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6\.venv\Lib\site-packages\starlette\testclient.py:51: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
  _PortalFactoryType = Callable[[], AbstractContextManager[anyio.abc.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 15 passed, 1 warning in 0.18s =====
Milestone command(s) completed.
```

Milestone 1 - Starter tests pass locally before modifications.

```
python -m pytest -v
```

October 6 revalidation of the unmodified starter commit 333bc5f in a separate checkout; the source state predates the assignment changes.

02 Baseline CI

jwm-dev / SDI_4213_Weeks_5_6

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<> Code

⌚ Issues

🔗 Pull requests

🔗 Agents

⌚ Actions

📅 Projects

📖 Wiki

More ⌵

← CI

✔️ Add Weeks 5-6 starter project #1

Re-run all jobs ⋮

🏠 Summary

All jobs ⌵

✔️ test-and-build

Run details

⌚ Usage

📄 Workflow file

Triggered via push 5 days ago

Status

Total duration

👤 jwm-dev pushed → 333bc5f main **Success** 25s

Artifacts

—

ci.yml

on: push

✔️ test-and-build 21s

🔗 - +

Annotations

1 warning and 1 notice

⚠️ test-and-build

Node.js 20 is deprecated. The following actions target Node.js 20 but are be...
[Show more](#)

ℹ️ test-and-build

"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 202...
[Show more](#)

Milestone 2 - Baseline GitHub Actions CI workflow passes.

https://github.com/jwm-dev/SDI_4213_Weeks_5_6/actions/runs/36925881257

Historic October 1 baseline run; the GitHub page was captured October 6.

Jeffrey Gregory | October 6, 2026 | Milestone 2

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03 Build Artifact

jwm-dev / SDI_4213_Weeks_5_6

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<> Code

🕒 Issues

🔗 Pull requests

🔄 Agents

🕒 Actions

📦 Projects

📖 Wiki

More

← CI

✔️ Add automated build artifact #2

Re-run all jobs

⋮

🏠 Summary

All jobs

✔️ test-and-build

Run details

🕒 Usage

📄 Workflow file

Triggered via pull request 5 days ago

Status

👤 jwm-dev opened #2 `feature/build-artifact`

Success

Total duration

Artifacts

52s

1

ci.yml

on: pull_request

✔️ test-and-build

13s

📄

−

+

Annotations

1 warning and 1 notice

⚠️ test-and-build

Node.js 20 is deprecated. The following actions target Node.js 20 but are be...

Show more

ℹ️ test-and-build

"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 202...

Show more

Artifacts

Produced during runtime

Name	Size	Digest
📦 sdi4213-app	3.66 KB	sha256: ab0121468072ae... 📄 ⬇️ 🗑️

Milestone 3 - CI created and uploaded the build artifact.

https://github.com/jwm-dev/SDI_4213_Weeks_5_6/actions/runs/36926050064

The successful test-and-build job uploaded sdi4213-app.

Jeffrey Gregory | October 6, 2026 | Milestone 3

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04 Artifact Contents

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 4 • Completed • exit 0

```
SDI 4213-980 | Weeks 5-6 | Jeffrey Gregory | Windows verification
Repository: https://github.com/jwm-dev/SDI_4213_Weeks_5_6
Captured session: 2026-10-06 09:00:37 -05:00

PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> Get-ChildItem dist/extracted

Mode                Length      Name
----                -
d-----            app
-a----      3995    README.md
-a----       88    requirements.txt
-a----        6    VERSION

PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> Get-ChildItem dist/extracted/app

Mode                Length      Name
----                -
-a----      1160    main.py
-a----       296    models.py
-a----      1236    services.py
-a----        63    __init__.py

PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> Get-Content dist/extracted/VERSION
0.1.0

PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> Get-FileHash dist/downloaded/sdi4213-app.zip -Algorithm SHA256

Algorithm : SHA256
Hash      : 3BDA490CA73B16A8E175398F323B83113D42A1A67DD8341BF83131A9A6FD1300

Milestone command(s) completed.
```

Milestone 4 - Downloaded artifact contains the required release files.

Get-ChildItem dist/extracted; Get-ChildItem dist/extracted/app; Get-Content dist/extracted/VERSION; Get-FileHash dist/downloaded/sdi4213-app.zip -Algorithm SHA256

Downloaded workflow ZIP contains app/, requirements.txt, README.md, and VERSION (0.1.0).

05 Version Tag and Release

jwm-dev / SDI_4213_Weeks_5_6

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

More

Releases / v0.1.0

Version 0.1.0 - Build and Containerization Exercise

Compare

jwm-dev released this 5 days ago

3 commits to main since this release

v0.1.0

3ce31ba

Version 0.1.0 of the instructor-provided FastAPI inventory application for SDI 4213 Weeks 5–6.

- Includes the existing inventory API and `/health` endpoint.
- Preserves all 15 automated pytest tests, verified locally and in GitHub Actions with Python 3.13.
- CI packages `app/`, `requirements.txt`, `README.md`, and `VERSION` into `sdi4213-app.zip` after tests pass, and uploads the package using `actions/upload-artifact@v4`.
- The attached ZIP was downloaded from the successful build PR workflow and checked for integrity, required file contents, and version `0.1.0`.

Build pull request: [#2](#)
Build workflow: https://github.com/jwm-dev/SDI_4213_Weeks_5_6/actions/runs/36926050064
Changelog: https://github.com/jwm-dev/SDI_4213_Weeks_5_6/blob/v0.1.0/CHANGELOG.md

Following the assignment's sequence, this tag records the Week 5 build release. The Week 6 Dockerfile and container evidence are delivered in a subsequent pull request on `main`; the Docker image uses the same application version, `sdi4213-week56:0.1.0`.

Week 6 completion:

- Docker pull request merged after CI and self-review: [#4](#)
- Final Docker PR workflow: https://github.com/jwm-dev/SDI_4213_Weeks_5_6/actions/runs/36926836443
- Complete submission evidence and reflections: https://github.com/jwm-dev/SDI_4213_Weeks_5_6/blob/main/docs/evidence-template.md
- Docker Desktop successfully built and ran `sdi4213-week56:0.1.0`. `/health` returned `{"status":"ok"}`, logs showed HTTP 200, and the image remained after the demo container was stopped and removed.

Assets 3

sdi4213-app.zip

sha256:3bda490ca...

3.92 KB

5 days ago

Source code (zip)

5 days ago

Source code (tar.gz)

5 days ago

Milestone 5 - Versioned GitHub Release v0.1.0 published.

https://github.com/jwm-dev/SDI_4213_Weeks_5_6/releases/tag/v0.1.0

The tag, release title, release notes, and attached application ZIP are visible.

Jeffrey Gregory | October 6, 2026 | Milestone 5

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06 Versioned Docker Image

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 6 • Completed • exit 0

```
SDI 4213-980 | Weeks 5-6 | Jeffrey Gregory | Windows verification  
Repository: https://github.com/jwm-dev/SDI_4213_Weeks_5_6  
Captured session: 2026-10-06 09:01:19 -05:00
```

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> docker images sdi4213-week56
```

Info -> U In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
sdi4213-week56:0.1.0	1cf7aad2feb	251MB	59.7MB	U

```
Milestone command(s) completed.
```

Milestone 6 - Versioned Docker image built successfully.

```
docker images sdi4213-week56
```

Windows Docker Desktop built image sdi4213-week56:0.1.0.

07 Running Container

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 7 • Completed • exit 0

```
SDI 4213-980 | Weeks 5-6 | Jeffrey Gregory | Windows verification
Repository: https://github.com/jwm-dev/SDI_4213_Weeks_5_6
Captured session: 2026-10-06 09:04:05 -05:00
```

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> docker ps --filter name=sdi4213-week56-demo
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
2fe4e125cf06	sdi4213-week56:0.1.0	"python -m uvicorn a..."	4 minutes ago	Up 4 minutes	0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp

sdi4213-week56-demo

```
Milestone command(s) completed.
```

Milestone 7 - Container is running and port 8000 is published.

```
docker ps --filter name=sdi4213-week56-demo
```

The demonstration container is running with host port 8000 mapped to container port 8000.

08 Health Check

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 8 • Completed • exit 0

```
SDI 4213-980 | Weeks 5-6 | Jeffrey Gregory | Windows verification  
Repository: https://github.com/jwm-dev/SDI_4213_Weeks_5_6  
Captured session: 2026-10-06 09:04:20 -05:00
```

```
PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> Invoke-RestMethod http://localhost:8000/health
```

```
status : ok
```

```
Milestone command(s) completed.
```

Milestone 8 - Containerized application health check succeeds.

Invoke-RestMethod http://localhost:8000/health

A successful response reports status ok. An earlier request raced startup; subsequent requests returned HTTP 200.

09 Container Logs

SDI_4213_Weeks_5_6

PowerShell live session • actual Windows command output

Milestone 9 • Completed • exit 0

```
SDI 4213-980 | Weeks 5-6 | Jeffrey Gregory | Windows verification
Repository: https://github.com/jwm-dev/SDI_4213_Weeks_5_6
Captured session: 2026-10-06 09:04:37 -05:00

PS C:\Users\jwm-d\Desktop\sdi-4213 weeks 5&6 assignment\SDI_4213_Weeks_5_6> docker logs sdi4213-week56-demo
INFO:      Started server process [1]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:      172.17.0.1:43000 - "GET /health HTTP/1.1" 200 OK
INFO:      172.17.0.1:60608 - "GET /health HTTP/1.1" 200 OK

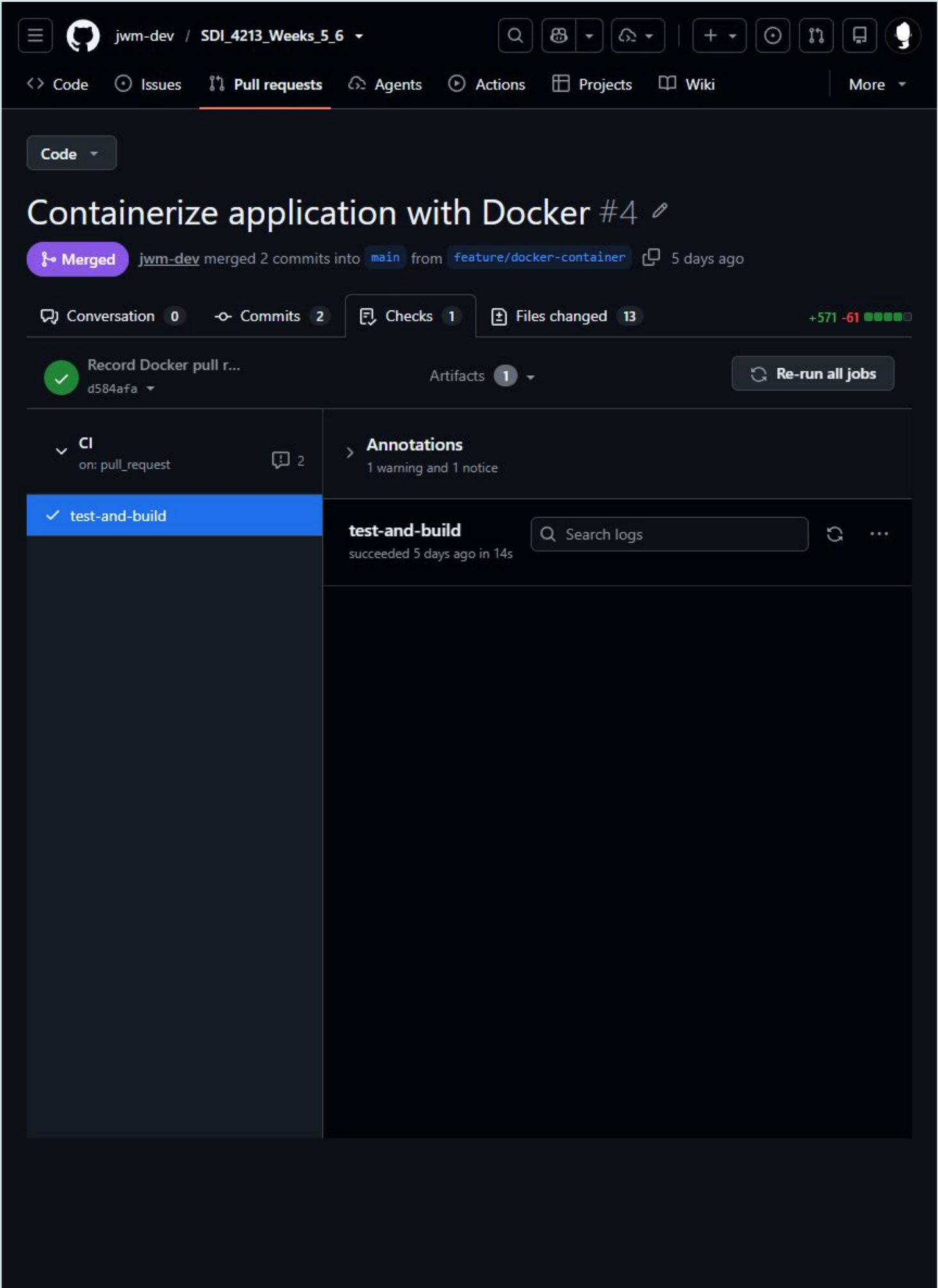
Milestone command(s) completed.
```

Milestone 9 - Container logs confirm the application started and handled requests.

```
docker logs sdi4213-week56-demo
```

Uvicorn starts on 0.0.0.0:8000 and successful /health requests appear in the logs.

10 Final Pull Request and CI



Milestone 10 - Containerization changes passed CI and were merged through the pull request workflow.

https://github.com/jwm-dev/SDI_4213_Weeks_5_6/pull/4

Historic merged Docker PR and passing checks. Final successful CI run:
https://github.com/jwm-dev/SDI_4213_Weeks_5_6/actions/runs/36926836443

Reflection

1. Build artifact, image, and container

A workflow artifact is a retained file package produced by CI. This ZIP contains application source, requirements, README, and version metadata; it still needs Python and installed dependencies to run. A Docker image packages the application, runtime, dependencies, filesystem layers, and startup command. A container is an instance created from an image. Removing the demonstration container leaves its image available for another run.

2. Version labels and repeatability

VERSION 0.1.0 identifies the application version. Git tag v0.1.0 points to a source commit, and its GitHub Release publishes notes and a downloadable artifact. Image tag sdi4213-week56:0.1.0 connects the runtime package with the same application version. Fixed release labels aid troubleshooting and rollback, while recorded commit and image IDs identify exact objects. Git and Docker tags can be moved, so version tags should be treated as immutable release labels.

Automation repeats the same test and packaging steps; the versioned image supplies the runtime and installed dependencies, reducing differences between environments. The Python slim base tag itself can change, so a production workflow should pin a base-image digest for stronger repeatability.

3. Port mapping

The option `-p 8000:8000` publishes host TCP port 8000 to container port 8000. A request to `http://localhost:8000/health` therefore reaches the containerized Uvicorn process, which listens on `0.0.0.0:8000`. The first number is the host port and the second is the container port. EXPOSE 8000 documents the intended port; the `-p` option publishes it.

4. Next automation toward deployment

After tests pass, build the Docker image in CI, scan it, run a container health check, and publish it to a registry with a version tag and recorded digest. Deploy that digest to staging, verify health, then promote the same image with an approval and rollback process. Store credentials in managed secrets and keep configuration external. Replace the application's in-memory inventory with persistent storage before relying on restarts or multiple replicas.

Supporting records

The completed docs/evidence-template.md, workflow, Dockerfile, .dockerignore, raw Windows verification logs, and ten source screenshots are available in the linked GitHub repository. Screenshots above preserve the captured application or terminal areas in numerical order.

https://github.com/jwm-dev/SDI_4213_Weeks_5_6